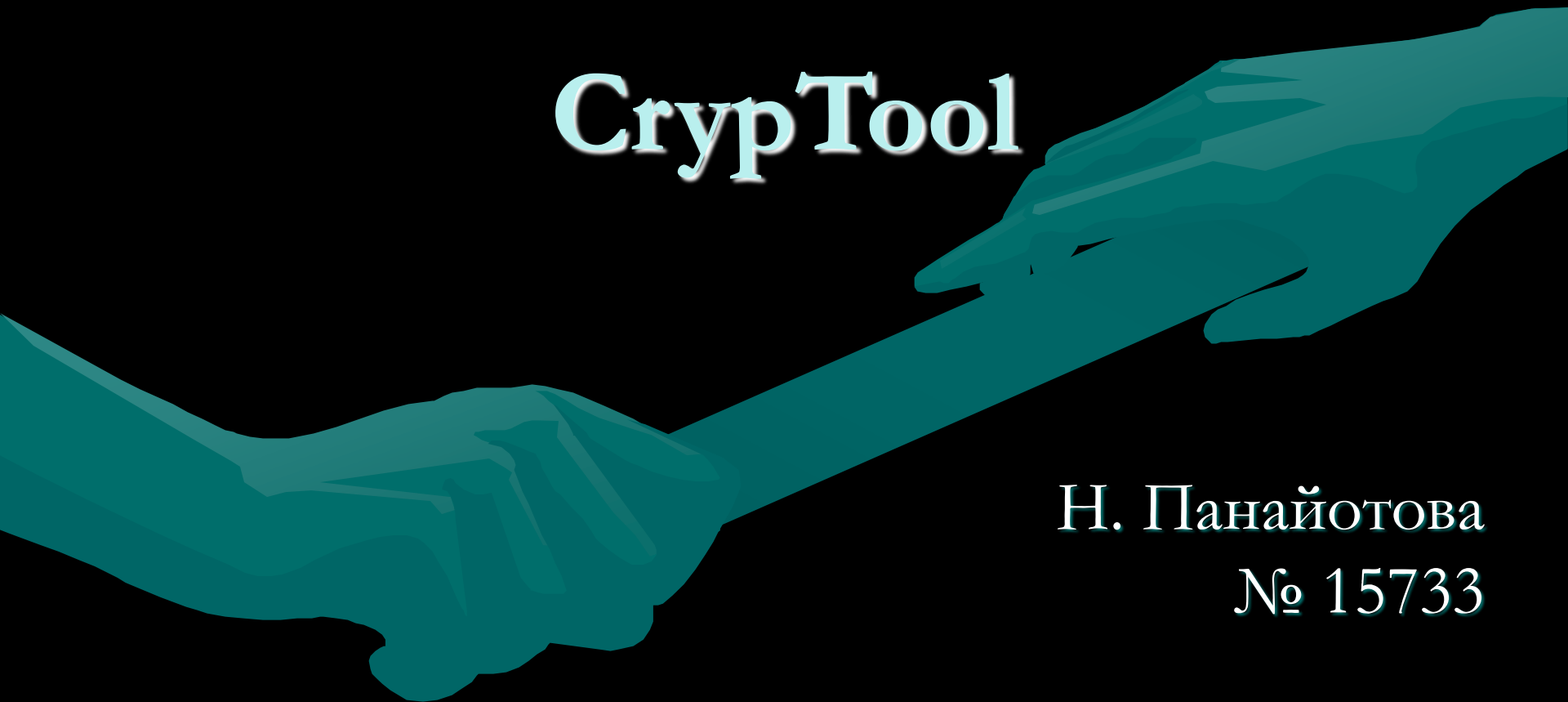


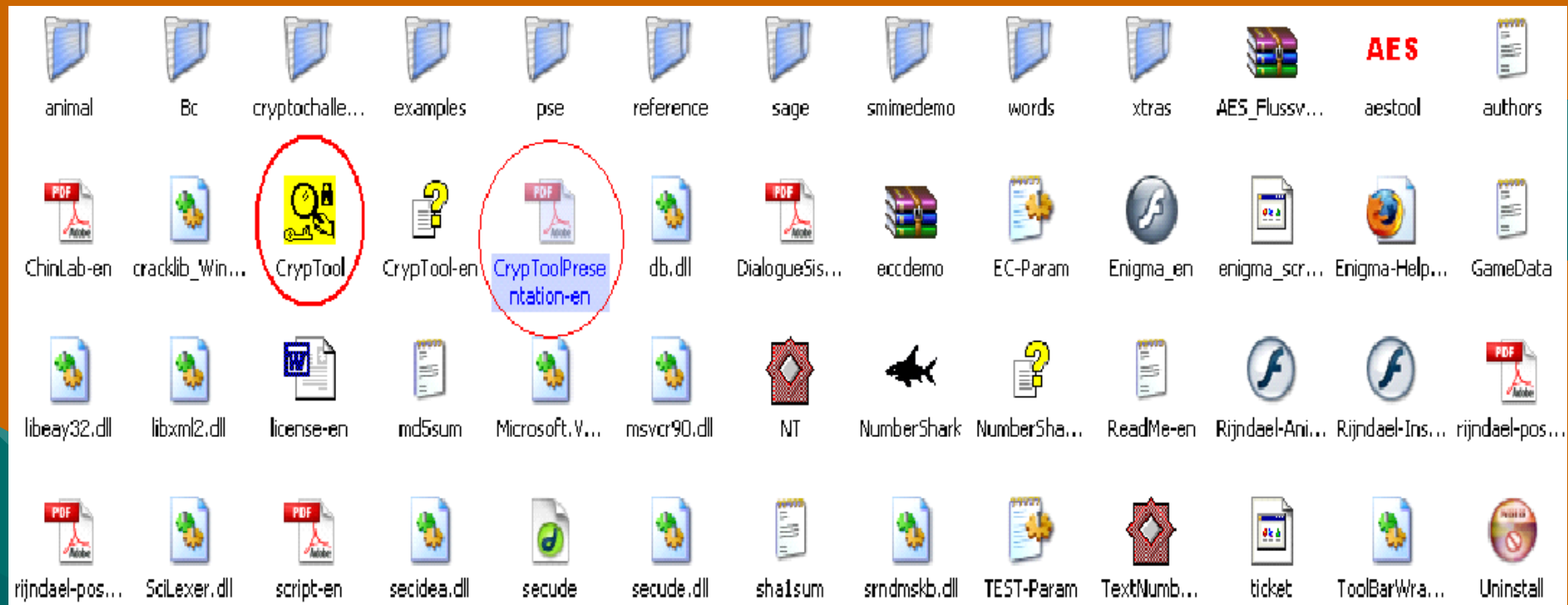


CrypTool



Н. Панайотова
№ 15733

C:\Program Files\Cryptool



Examples of Early Cryptography (2)

Caesar encryption (mono-alphabetic substitution cipher)

- **Caesar encryption** (Julius Caesar, 100 - 44 B.C.)
- Simple substitution cipher

GALLIA EST OMNIS DIVISA ...

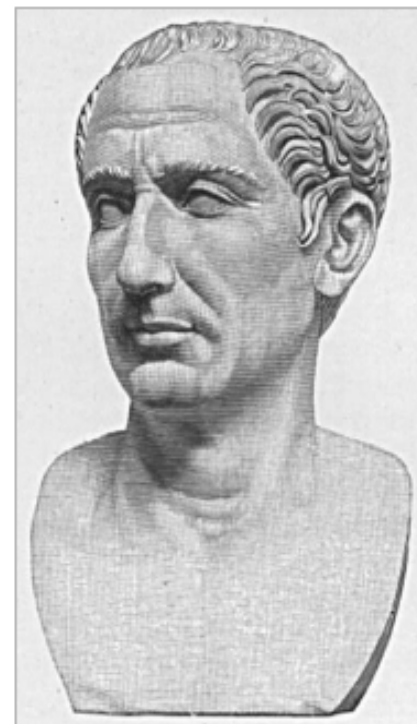
Plaintext: ↘

ABCDEF**G**H IJKLMN O PQRSTU VWXYZ

Secret alphabet: ↓

DEFGH**I**JKLMN O PQRSTU VWXYZABC

↙ **J**DOOLD HVW RPQLV GLYLVD ...



- **Attack: Frequency analysis** (typical character allocation)

Presentation with CrypTool via the following menus:

- Animation: "Indiv. Procedures" \ "Visualization of algorithms" \ "Caesar"
- Implementation: "Crypt/Decrypt" \ "Symmetric (classic)" \ "Caesar / Rot-13"

Cesar/Rot-13

The screenshot displays the CrypTool 1.4.30 application window. The menu bar includes File, Edit, View, Encrypt/Decrypt, Digital Signatures/PKI, Individ. Procedures, Analysis, Options, and Windows. The 'Encrypt/Decrypt' menu is open, showing options like Symmetric (classic), Caesar / Rot-13..., Symmetric (modern), Hill..., Asymmetric, Substitution / Atbash..., Hybrid, Playfair..., ADFGVX..., Byte Addition..., XOR..., Vernam / OTP..., Homophone..., Permutation / Transposition..., Solitaire..., and Scytale / Rail Fence... The 'Caesar / Rot-13...' option is selected. A secondary window titled 'Key Entry: Caesar / ROT-13' is open, showing the 'Text Options' tab. This tab includes checkboxes for 'Keep characters not present in the alphabet unchanged', 'Distinguish between uppercase and lowercase', and 'Define the alphabet used in text ciphers'. The 'Alphabet to use' field contains 'ABCDEFGHIJKLMNOPQRSTUVWXYZ 0123456789abcdefghijklmnopqrstuvwxyz'. The 'Apply' button is highlighted. A third window titled 'ROT 13 encryption of <Unnamed1>, key <j, KEY OFFSET: 0>' shows the encrypted output: 'IIPG:FFC+\$J+#+=FDGI"\$EJ\$M"+&I""+"L=#<\$FE#C+GIF%I#D+#~FLK+=IPGKF%I#G\$P+#E\+=IPGK#E#CPJ\$JF&&"I\$E%+"OK"EJ\$M"+FEC\$E"+\$"CG+#E\+D#EP+M\$JL#C\$Q#<\$FEJR'.

I CrypTool is a comprehensive free educational program about cryptography and cryptanalysis offering extensive online help and many visualizations.

II Symmetric (classic)

III Caesar / Rot-13...

IV Rot-13

V Text options

VI Apply

VII Encrypt

VIII ROT 13 encryption of <Unnamed1>, key <j, KEY OFFSET: 0>

IIIPG:FFC+\$J+#+=FDGI"\$EJ\$M"+&I""+"L=#<\$FE#C+GIF%I#D+#~FLK+=IPGKF%I#G\$P+#E\+=IPGK#E#CPJ\$JF&&"I\$E%+"OK"EJ\$M"+FEC\$E"+\$"CG+#E\+D#EP+M\$JL#C\$Q#<\$FEJR

Solitaire



Solitaire De-/Encryption

Description

With this dialog you can make use of the Solitaire cipher. Usually 54 cards are used.

The different cards are represented as numbers from 1 to 52 and as letters A and B. The values of the different cards can be found in the online help (F1) at "Solitaire description".

The application of the Solitaire cipher is carried out in 3 main steps (the first and the second are preparatory steps; in the third step the encryption is realized):

Step 1: Either you create the key (the shuffled card deck) by yourself or you load a key (depending on your choice in sector "1 b: Defining the card order").

The result of generating the key is the shuffled card deck, which can be seen at "Initial deck" in the grouping "Output".

Step 2: You generate the keystream (This is the actual Solitaire algorithm). This process modifies the card deck.

Its actual card order can be found at "Final deck" in the grouping "Output".

Step 3: You encrypt the chosen text (In doing so each plaintext letter is XOR-added to each keystream letter).

In the case the chosen text is already encrypted with the Solitaire, you can also decrypt it.

Shuffling cards and generating keystream

1. Generating the initial deck (= shuffling) == Generating the key

1 a: Number of cards (classic: 52+2)

Total number of cards
(incl. joker):

54

Optional

Enter password

Modify the initial deck

The initial deck can be modified by using a password. You can read more about the functionality of the password in the online help.

Own Choice

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	A	B

Reset

1 b: Defining the card order

randomly shuffled

Different shuffle

2. Generating the keystream (automatically or manually)

Automatic

Manual

Manual steps of the Solitaire algorithm

Step 1

Step 2

Step 3

Step 4

Step 5

You still need

of

keystream letters.

Generate rest

Output of the different states of the card deck and of the generated keystream

Initial deck (already shuffled card deck; represents the state of the card deck before the algorithm to generate the keystream is used the first time)

A,52,51,49,6,12,48,13,18,7,11,17,28,3,10,21,14,19,16,20,23,22,29,15,24,38,25,35,9,30,32,34,41,31,42,27,39,33,26,44,36,2,43,50,8,40,5,45,47,37,8,4,46,1

Save

Final deck (represents the state of the card deck directly after generating the last keystream letter)

14,50,1,20,8,44,41,5,11,52,32,26,10,12,37,18,35,25,4,8,40,6,7,23,24,39,2,21,16,30,38,13,34,48,19,27,51,17,9,45,33,3,28,42,43,46,31,49,15,36,47,A,22,29

Save

Keystream

52,5,40,1,17,52,14,1,34,22,19,24,4,32,13,2,6,2,31,16,30,39,47,1,16,19,14,7,3,36,26,39,19,14,17,12,14,14,15,37,36,11,42,34,20,45,3,7,3,35,50,4,40,39,39,9,26,28,16,6,45,19,27,37,52,9,24,44,46,16,52,34,50,12,12,45,21,46,29,29,35,11,11,6,10,27,12,39,43,5,28,46,23,19,3,25,27,7,4,12,33,48,48,39,45,48,18,27,14,36,26,45,13,38,23,8,20,34,33,29,41,47,29,11,35,17,37

Save

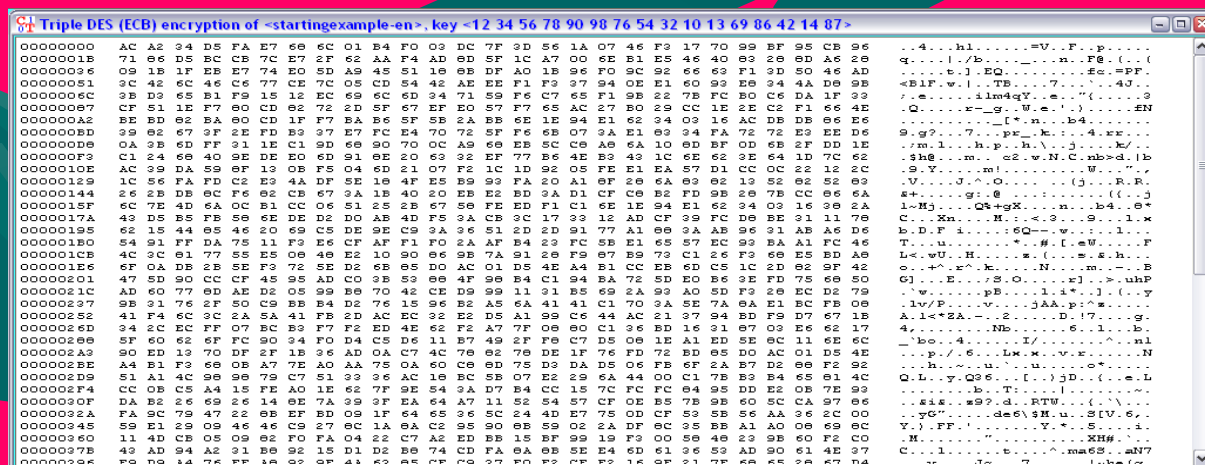
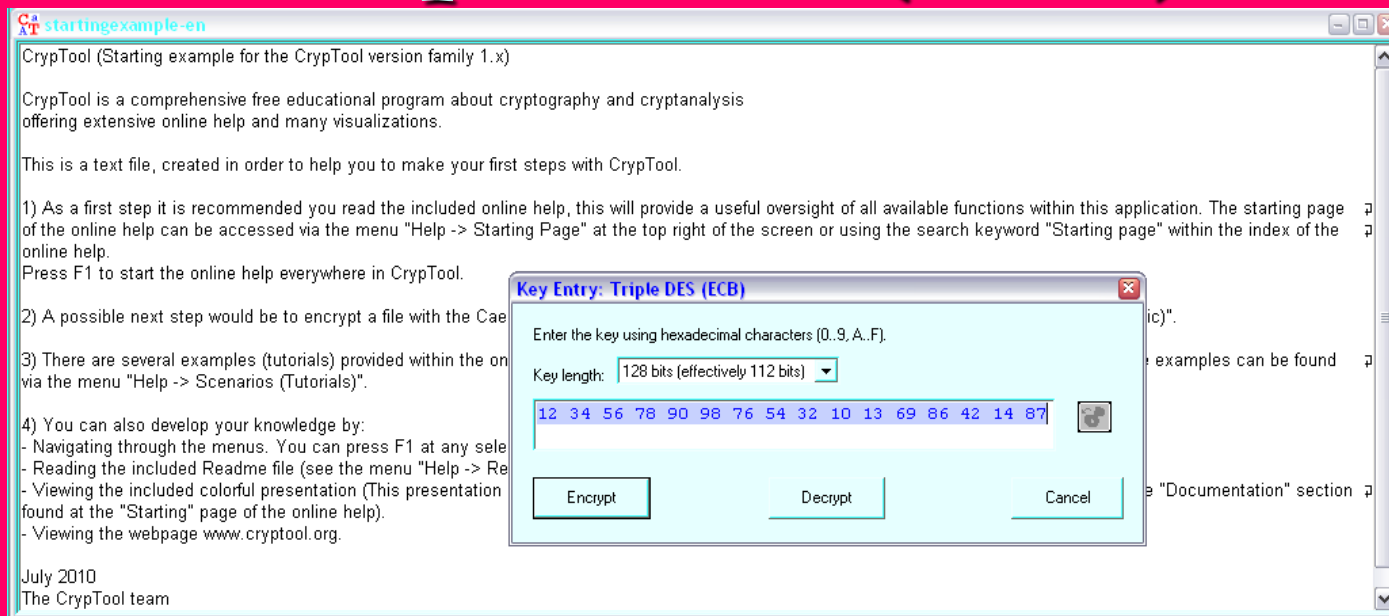
Encrypt

Decrypt

Generate initial state

Cancel

Triple DES (ECB)



Generate key for Digital Signatures/PKI

Generation of Asymmetric Key Pair

Algorithm

☒ RSA
Bit length of RSA modulus: 1024

☐ DSA
Bit length of DSA prime number: 1024

☐ Elliptic curves
Identify curve: []

User data

The key pair will be put in an encrypted PSE with the name shown below. The key pair will be protected by your PIN code.

Last name: Panayotova

First name: Nikoleta

Key identifier: (optional) []

SECUDE Crypto Runtime - Random Number Generator

Random Number Generation

Move your mouse and press different keys on your keyboard until enough random material is collected.

[Progress bar]

OK

The domain []

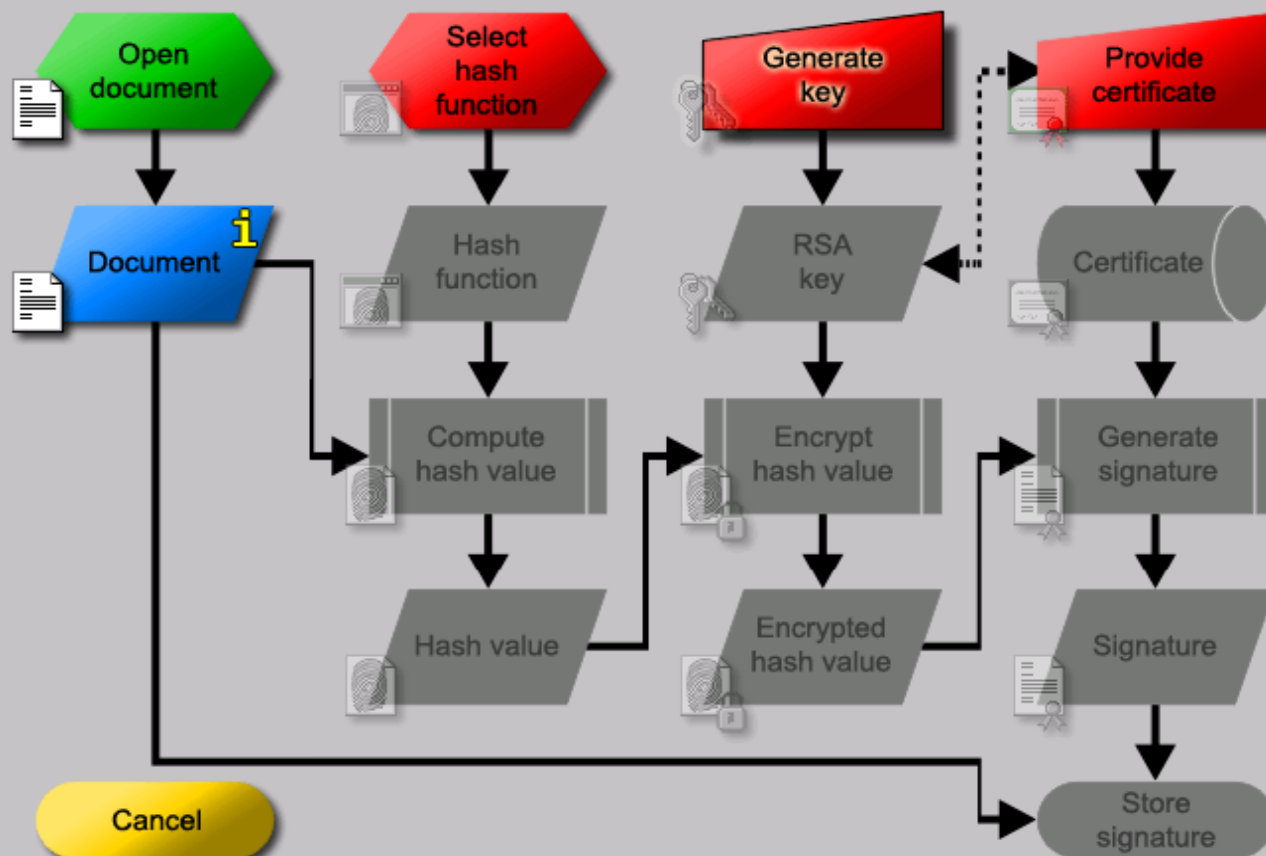
Parameter []

Base for presentation of numbers

☐ Octal ☒ Decimal ☐ Hexadecimal

Generate new key pair... PKCS #12 Import Show key pair... Close

Step by Step Signature Generation



Document: Triple DES (ECB) encryption of <startingexample-en>, key <12 34 56 78 90 98 76 54 32 10 13 69 86 42 14 87>

00000	AC	A2	34	D5	FA	E7	68	6C	01	B4	F0	03	DC	7F	3D	56	1A	07	~g4Xvshl.rp.b.=V..
00012	46	F3	17	70	99	BF	95	CB	96	71	86	D5	BC	CB	7C	E7	2F	62	Fy.p.i.II.q.XjJ s/b
00024	AA	F4	AD	8D	5F	1C	A7	00	6E	B1	E5	46	40	83	28	8D	A6	28	@-_.S.ntieF@.(. (
00036	09	1B	1F	EB	E7	74	E0	5D	A9	45	51	18	8B	DF	A0	1B	96	F0	...xstaj@EQ..Я..p
00048	9C	92	66	63	F1	3D	50	46	AD	3C	42	6C	46	C6	77	CE	7C	05	..fcc=PF-<B1F*W0 .
0005A	CD	54	42	AE	EE	F1	F3	37	94	0E	E1	60	93	E8	34	4A	D8	9B	HTB@ocy7..8`u4JW.
0006C	3B	D3	65	B1	F9	15	12	EC	69	6C	6D	34	71	59	F6	C7	65	F1	;Yeiw..miln4qYu3ec
0007E	9B	22	7B	FC	B0	C6	DA	1F	33	CF	51	1E	F7	80	CD	82	72	2D	..{b*%b.3PQ.v.H.r-
00090	5F	67	EF	E0	57	F7	65	AC	27	B0	29	CC	1E	2E	C2	F1	66	4E	_gnaWqe~'^)M..BcfN
000A2	BE	BD	82	BA	80	CD	1F	F7	BA	B6	5F	5B	2A	BB	6E	1E	94	E1	sS.e.H.ve!_[*>n..8
000B4	62	34	03	16	AC	DB	DB	86	E6	39	82	67	3F	2E	FD	B3	37	E7	b4...bM.*9.g?.ei7s

The math inside ☺

Prime Number Generation

Prime numbers play an important role in modern cryptography. Here you can generate primes within a given value range [lower limit, upper limit].

Amount of prime numbers to be generated

☒ Generate two primes randomly from within the value range(s)

☐ Generate all primes within the value range set for p

Separator for the display of the primes:

Algorithms for prime number generation

☐ Miller-Rabin Test

☐ Solovay-Strassen Test

☒ Fermat Test

Value range of the prime numbers p and q

☒ To be entered independently of each other

☐ Both are equal (just enter one)

Prime number p

Lower limit

Upper limit

Result

Prime number q

Lower limit

Upper limit

Result

Generate prime numbers

Cancel

Indiv. Procedures

Planetary Motion and the Chinese Remainder Theorem

The Chinese Remainder Theorem (CRT) can be used to solve systems of linear modular equations. Enter up to 9 equations $x = a[i] \bmod m[i]$ ($i=1, \dots, 9$) below. Such a system of equations can be used to determine the number of days until certain planets become aligned.

Simultaneous congruences / linear modular equations

$x \equiv$	15	mod	88
$x \equiv$	0	mod	225
$x \equiv$	100	mod	365
$x \equiv$	0	mod	687
$x \equiv$	0	mod	4327
$x \equiv$	0	mod	10751
$x \equiv$	0	mod	30661
$x \equiv$	0	mod	60149
$x \equiv$	0	mod	90491

Solution

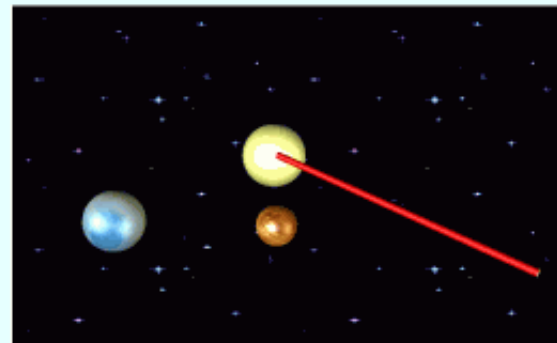
Solve

Exit

Clear all parameters

Restore default settings

Astronomical visualization



The periods of the planets Mercury and Earth around the Sun are 88 and 365 days, respectively. If it will take the planets

and days,

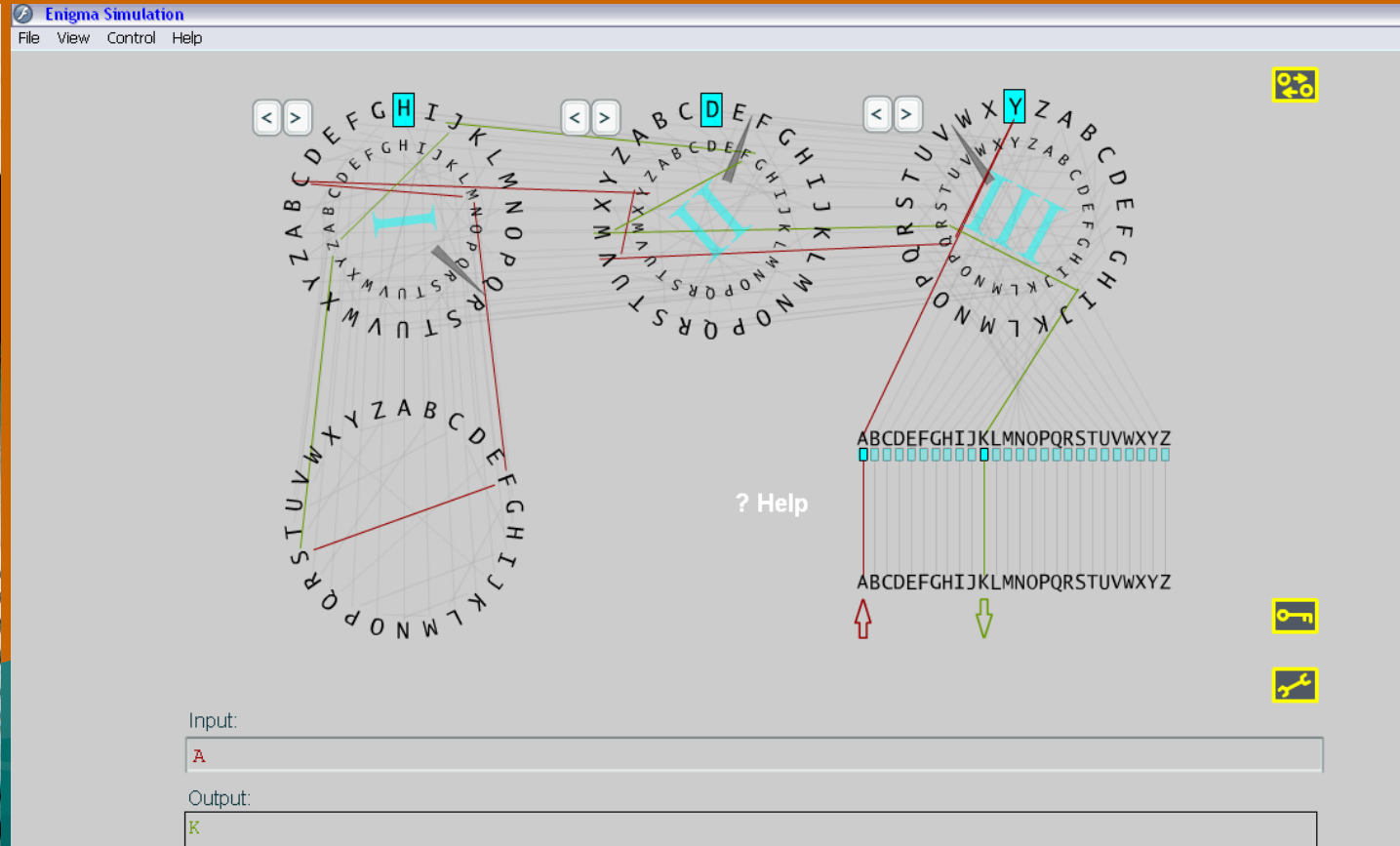
respectively, to reach a given radius vector s (in red), is it possible that the planets will ever lie simultaneously on s ?

Choose a planet

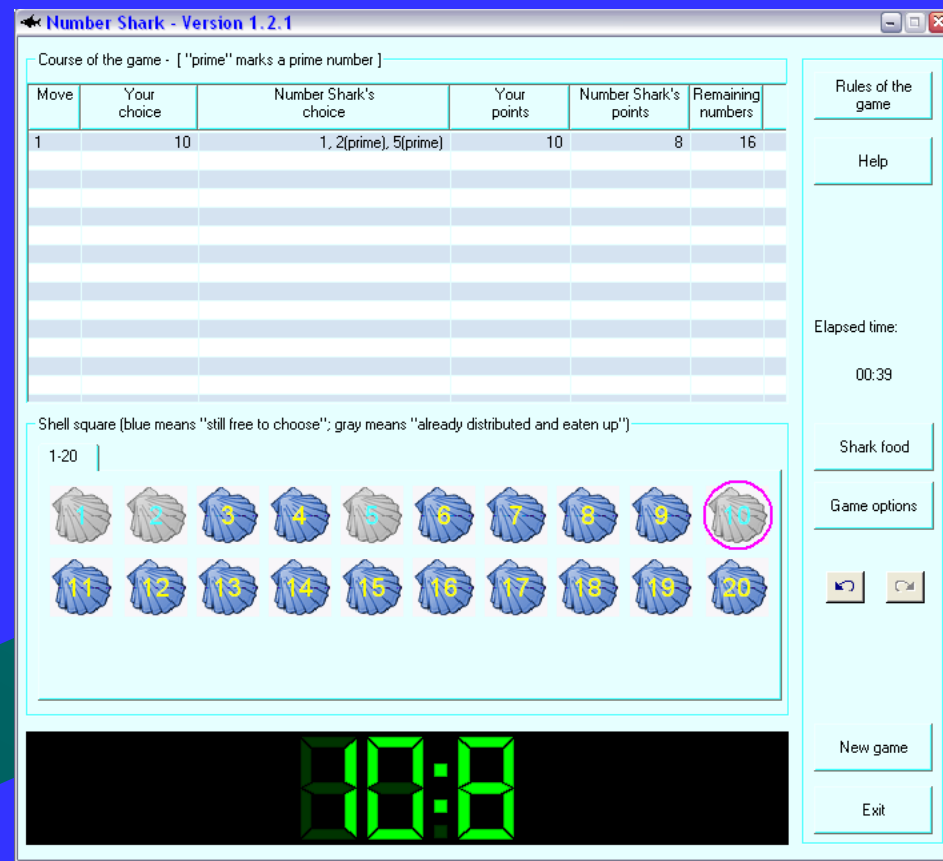
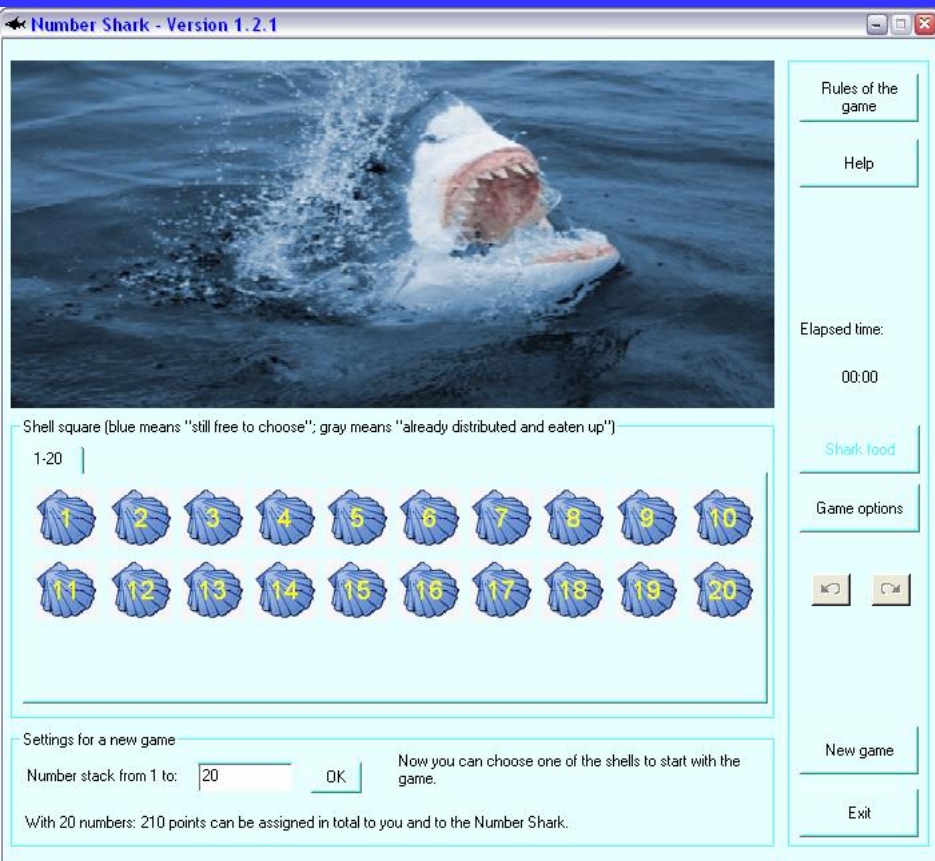
- | | | |
|---|---|---|
| ☒ Mercury | ☒ Mars | ☒ Uranus |
| ☒ Venus | ☒ Jupiter | ☒ Neptune |
| ☒ Earth | ☒ Saturn | ☒ Pluto |

In what time interval (in days) will this incident repeat itself?

Visualization of Algorithms/Enigma



Number Shark



New game?













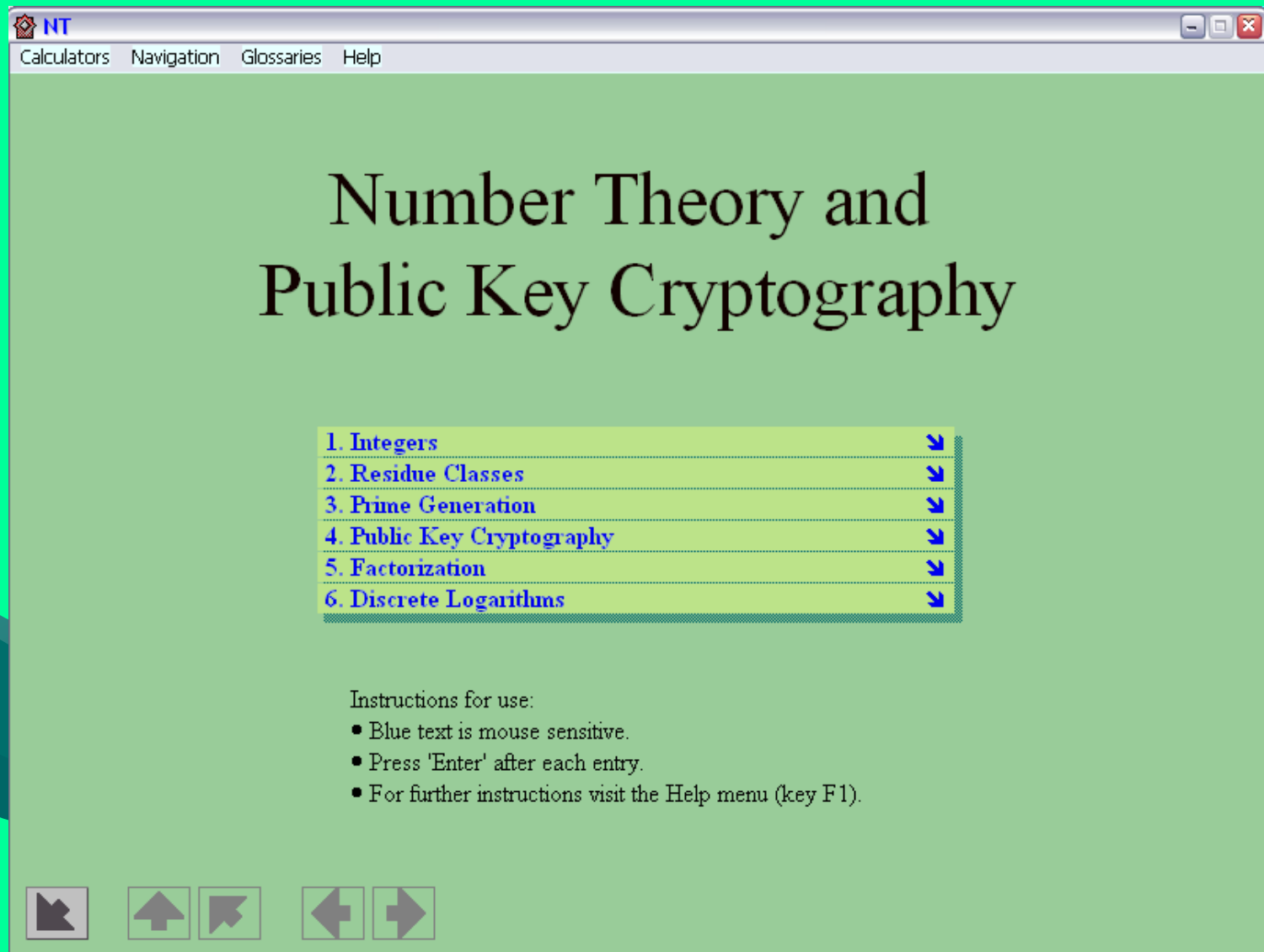








On dry land!!!





1.2 Primes

page 5 of 21

Definition: A **prime** or prime number p is a natural number with exactly two positive divisors, namely 1 and p itself. The set of all primes is called **IP**.

Notice that 1 is no prime. The primes < 100 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Definition: If a prime p divides an integer a , then p is called **prime divisor** of a .

Theorem: Each natural number $n > 1$ has at least one prime divisor. [Proof](#)

From this [Euclid](#) got insight in the number of primes:

Theorem: There are infinitely many primes. [Proof](#)

Definition: The natural numbers between the primes are called **composite numbers**.

Theorem: Each composite number n has a prime divisor $p \leq \sqrt{n}$. [Proof](#)

Thus, if an integer $n > 1$ is not divisible by any prime $p \leq \sqrt{n}$, then n is prime.

Example: 89 is prime, since 2, 3, 5, 7 do not divide 89, and $11 > \sqrt{89}$.



(Go on to the next page.)

3.2 Fermat Test

page 4 of 11

Each prime p passes a test that results from Fermat's [Little Theorem](#):

For $b \in \{2, \dots, p-1\}$, test if $b^{p-1} \equiv 1 \pmod{p}$.

This test is called **Fermat Test**. Unfortunately some composite numbers pass it as well.

Example: $341 = 11 \cdot 31$, and yet $2^{340} \equiv 1 \pmod{341}$.

Passing the test provides no information. It must be repeated with a different base b :

$n =$ $2^{n-1} \equiv 1 \pmod{n}$ Test passed
 $\text{GCD}(b, n) = 1$ b

Definition: Let n be a composite number coprime to b .

If $b^{n-1} \equiv 1 \pmod{n}$, then it is said that

- n is **pseudoprime to b** ,
- and • b is a **liar for** (the primality of) n ,

otherwise it is said that

- b is a **witness against** (the primality of) n .

Theorem: If there are any witnesses against n ,
 then they make up at least 50% of all $b \in \{1, \dots, n\}$ coprime to n . [Proof](#)



(Go on to the next page.)

Other versions:



CrypTool 2.0 Beta 02 (2.0.3577) - Present-Sample-01.cte (Running)

Home Settings Algorithms Help

New Open Save Play Pause Stop Undo Redo Add workspace Remove workspace Current workspace 1/1 Import subworkspace

CrypTool workspace Process the elements in the workflow Plugin editor

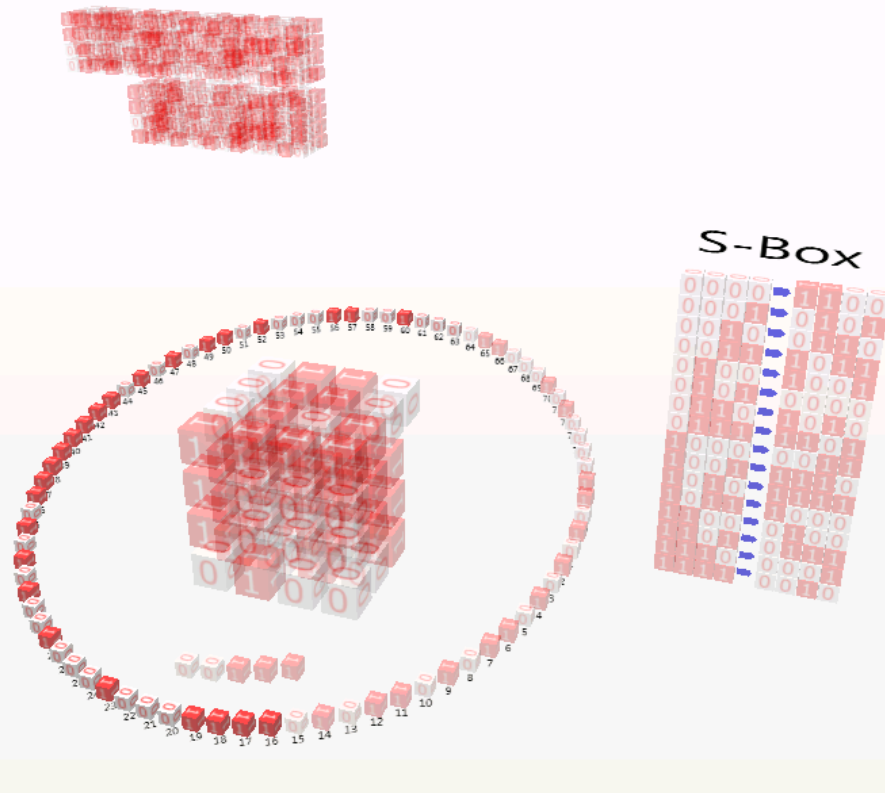
- Algorithms
- Modern Ciphers**
- Symmetric (Block and stream)
- AES
 - DES
 - HIGHT
 - PRESENT**
 - Rivest Cipher / Ron's Code
 - SDES
 - TEA
 - Triple DES
 - Twofish
 - WEP protocol
- Asymmetric
- RSA
 - RSAGenerator
- Hybrid
- Misc

- Classic Ciphers
- Modern Ciphers**
- Hash Functions
- Cryptanalysis
- Tools
- Editor Specific

Help Present-Sample-01.cte PRESENT.enc (Presentation)

About Settings Overview Key Schedule Encryption Trace

picking the 64 left-most bits as roundkey 7 (0x61461316aff5488f)



Key Schedule Round: 07 Step: Key Extraction

13 Messages (51 filtered)

0 Errors 0 Warnings 13 Infos 51 Debugs Delete all messages Export to HTML

Nr	LogLevel	Time	Plugin	Title	Message
61	Info	15:12:25:734	PRESENT	PRESENT.dec	Decryption complete! (in: 80 bytes, out: 75 bytes)
62	Info	15:12:25:734	PRESENT	PRESENT.dec	Wrote data to file: C:\Dokumente und Einstellungen\Arno Wacker\Anwendungsdaten\CrypTool Temp Files\48de14a-9cdc-4b92-a188-d611fdfdb2db\41363d13-340d-4af2-baca-522524e9e820.csf

Algorithm Settings

PRESENT.enc

PRESENT is an ultra-lightweight block cipher

Show plugin description

Action: encrypt

Chaining Mode: ECB

Padding Mode: PKCS7

JCrypTool

JCrypTool

The logo for JCrypTool, featuring the text "JCrypTool" in a white, sans-serif font. To the right of the text is a stylized graphic consisting of a series of horizontal lines that curve and end in three arrows pointing to the right.

The cryptography e-learning platform.

Getting started with JCrypTool

May 2009 edition

Thank you! ;)

